Gabriel Becker

New Orleans, LA • gabebecker2048@gmail.com • gabebecker.com • github.com/GabeBecker2048

SUMMARY

Data Engineer with 2+ years owning production data pipelines end to end: ingesting, cleaning, reconciling, and standardizing ~200 financial-market datasets (100+ TB) into one trusted platform. Builds production ETL in Python and SQL on Apache Airflow, ships through GitLab CI with automated linting, type-checking, and tests, and runs Docker-based environments across staging and production. Led research into using agentic AI to generate pipelines and onboard new datasets. Builds and publishes open-source Python software outside of work.

TECHNICAL SKILLS

- **Languages:** Python (8+ years), SQL, C/C++ (academic), JavaScript and R (basic)
- **Data Engineering:** Apache Airflow, ETL/ELT pipeline design, data modeling, Apache Parquet, Apache Kafka, Spark / Databricks (basic)
- **Python Data Stack:** pandas, Polars, PyArrow, NumPy, SciPy, DuckDB
- **Databases & APIs:** PostgreSQL, SQLite, DuckDB, Parquet data lakes; FastAPI, REST, WebSockets
- **Cloud & Containers:** AWS (S3, CLI), Docker, Docker Compose, Linux
- **CI/CD & Testing:** GitLab CI, GitHub Actions, pytest, mypy, black, ruff, automated build/publish/deploy pipelines
- **Version Control:** Git: branching, merge requests, code review

PROFESSIONAL EXPERIENCE

Code Willing Inc, New Orleans, LA — Data Engineer *Apr 2024 – Present*

- Own the ingestion of third-party financial data: ~200 datasets and 100+ TB of equities, futures, options, and FX data, cleaned, reconciled, and merged into a uniform schema.
- Design, build, and maintain production Apache Airflow ETL pipelines in Python and SQL, from daily schedules to 5-minute intervals and event-driven runs triggered as source files land.
- Enforce data-quality standards end to end, catching and resolving source inconsistencies, schema drift, and delivery errors before they reach downstream users, under strict delivery deadlines.
- Lead developer of a Parquet conversion tool that replaced an existing tool, running up to 5× faster on one-half to two-thirds the memory; standardized every dataset to a consistent directory structure, format, and schema.
- Primary maintainer of the team's core pipeline repository and its GitLab CI/CD: automated black, mypy, and pytest stages on every merge request, a build stage that packages a wheel and publishes docs to S3, and a scripted deployment into production environments.
- Solely designed, built, and maintained a Docker Compose staging Airflow environment, giving a 10-person data team pre-production DAG testing where pipelines were previously tested in production.
- Led internal research into agentic AI (Claude Code), producing a workflow that generates complete pipelines and onboards new datasets with minimal manual coding.

De La Salle High School, New Orleans, LA — Mathematics Teacher *Oct 2023 – Dec 2023*

- Taught two sections of Algebra 2 (40+ students); wrote all lesson plans and assessments and reported progress to administrators and parents.

Delgado GenCyber Camp (Go Learn Labs), New Orleans, LA — Assistant Instructor *May 2023 – Jul 2023*

- Supported instruction at a month-long cybersecurity camp for 40+ high school students.

PROJECTS

Demos, documentation and more projects at gabebecker.com/projects

- **Murmur (Python, FinBERT, SQLite, Parquet, FastAPI):** Lead developer of a daily pipeline that scores news-headline sentiment for the 100 largest US companies, stores it beside their prices, and tests whether news leads, lags, or moves with returns. Idempotent SQLite store, Parquet exports, a scheduler and job queue, and a REST API with a dashboard.
- **chesssnake (Python, FastAPI, PostgreSQL, WebSockets, Docker):** A pure-Python chess engine, board renderer, and REST game server on SQLite or PostgreSQL, with live updates over websockets and a web UI. Published on PyPI and as a Docker image.
- **anchorless (Python, ffmpeg, Docker):** Turns public RSS feeds into a 24/7 news TV channel for Jellyfin, Plex, and IPTV players, grouping the same story across outlets and interrupting for breaking news. Published on PyPI and as a Docker image.
- **how-much-music (Python, JavaScript):** An exact count of every possible 4/4 measure on a piano (about 2.08×10^{35} written forms), derived as a recurrence and published as a paper with an interactive calculator.

EDUCATION

Mississippi State University, Starkville, MS *Aug 2019 – May 2023*

B.S. in Computer Science, Concentration in Artificial Intelligence; Minor in Mathematics. President's Scholar; Dean's Scholar (3 semesters).
Coursework: Linear Algebra, Multivariable Calculus, Differential Equations, Engineering Statistics, Algorithms, Machine Learning.
Study abroad: Comillas Pontifical University, Madrid, Spring 2022.